

Neuroeconomic Foundations of Familiarity Bias: An Integrated Analysis of Multi-dimensional EEG Indices and Risk Preference Parameters

Abstract

This study employs a neuroeconomic approach to investigate the neural mechanisms underlying familiarity bias in risky decision-making and their association with risk preference parameters. While familiarity bias is prevalent in financial markets and consumer choices, traditional economic models struggle to explain its underlying mechanisms. We integrate ERP and neural oscillation analyses to provide evidence of the neural basis of familiarity bias and examine its correspondence with economic risk preference parameters.

We implemented a city temperature-based lottery paradigm using a 3 (risk source: Beijing/familiar, Tokyo/unfamiliar, random numbers/neutral) $\times$ 2 (frame: gain/loss) within-subject design. Participants (N=31) completed 162 binary choice trials while behavioral data and EEG activity were recorded simultaneously. Behavioral data were used to estimate CRRA risk preference parameters, while neural data were analyzed for P300 and N200 components, as well as time-frequency dynamics in Delta (1-4Hz) and Theta (4-8Hz) bands.

Behavioral results show that unfamiliar conditions significantly increased risk-seeking behavior and prolonged reaction times, indicating that familiarity influences both subjective value assessment and cognitive processing efficiency. The interaction between framing effects and familiarity, with more pronounced framing effects under unfamiliar conditions, is not entirely consistent with prospect theory predictions, suggesting that familiarity may modulate the shape characteristics of the value function.

ERP analysis revealed that unfamiliar conditions in the gain frame elicited larger P300 amplitudes ($p=.054$) with no significant differences in N200 ($p=.132$), indicating that familiarity primarily affects later evaluation processes rather than early information encoding in gain contexts. Conversely, familiar conditions in the loss frame elicited significantly larger P300 ($p=.047$) and N200 amplitudes ($p=.036$), revealing that

preference for familiar information sources in loss contexts may begin at early risk assessment stages. Time-frequency analysis found that Delta oscillations showed frame sensitivity in early (0-0.3s) and middle (0.3-0.6s) time windows under unfamiliar conditions ($p < .09$), while Theta oscillations demonstrated significant framing effects under unfamiliar conditions ($p = .002$) and familiarity modulation in early stages of loss framing ($p = .008$).

These neural findings suggest that the economic significance of familiarity bias manifests in two stages of risk decision-making: early risk source evaluation (corresponding to N200 and Theta oscillations) may influence subjective probability weights in expected utility theory, while later value integration processes (corresponding to P300 and Delta oscillations) directly relate to risk attitudes measured by CRRA. Particularly, the neural activity pattern for familiar conditions under loss framing aligns with classic loss aversion phenomena, suggesting that familiarity may modulate risk preferences by adjusting risk probability assessment rather than merely altering subjective value.

By linking neural indices with economic parameters, this study provides new insights into the formation mechanisms of familiarity bias, offering integrated evidence for behavioral economics and neuroeconomic theory development. These findings complement traditional economic understanding of decision biases and have potential implications for financial decision-making, consumer behavior, and policy design.